ORIGINAL RESEARCH

Reliability of Anthropometry-Based Equations Compared to Dual Energy Absorptiometry for Assessing Body Composition in Predialysis Chronic Kidney Disease—A Longitudinal Study

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Objectives: Skinfold thickness measurements for assessing body composition are reported to have good reproducibility compared to the reference method of dual energy absorptiometry (DXA). In the current study, we compared the level of agreement between body composition measured with DXA and skinfold thickness (SFT) in CKD Stage 3 and 4, at 2 occasions, 6 months apart.

Methods: Body composition was assessed in 177 Indian patients with CKD Stage 3 and 4 using DXA and anthropometry (SFT). The body fat mass obtained by the 2 methods was compared by paired *t*-test, intraclass correlation coefficients, regression analysis, and Bland-Altman plots. A linear regression analysis was done to identify the patient-related parameters which would account for the intermethod differences between DXA and SFT.

Results: Compared to DXA, SFT underestimated the fat mass at baseline as well as 6 months [DXA vs. SFT at entry: 15.85 kg (95% confidence interval, CI 15.07-16.65) vs. 13.71 kg (95% CI 13.21-14.32), $P < .001$; at 6 months: 16.13 (95% CI 15.33-16.93) vs. 13.85 (95% CI 13.25-14.45), $P < .001$]. The intraclass correlation coefficients at entry and 6 months were 0.894 (0.857-0.921) and 0.896 (0.860-0.923), respectively. The intermethod differences between DXA and SFT at baseline and 6 months were comparable: 2.08 kg (95% CI 1.66-2.5) at baseline versus 2.27 kg (95% CI 1.83-2.71) at 6 months, $P = 0.200$. Gender and body mass index turned out to be the significant predictors of intermethod differences at base line and exit ($P < .001$).

Conclusions: SFT-based measurements show good reproducibility compared to DXA over a period of 6 months. However, SFT systematically underestimates the fat mass by 2 Kg compared to DXA.

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Introduction

CHRONIC KIDNEY DISEASE (CKD) is associated with alterations in body composition. There is a higher prevalence of obesity as well as loss of muscle mass in patients with CKD. With declining kidney function, the prevalence of protein-energy wasting (PEW) and sarcopenia increases.[1] PEW is associated with higher mortality rates and morbidity in predialysis CKD.[2] Early identification of a poor nutritional status and introduction of appropriate remedial interventions are imperative in preventing complications, and improving morbidity and mortality in subjects with predialysis CKD. Conventional parameters like weight and body mass index (BMI) are influenced by altered hydration status and might not be an appropriate tool for assessing and monitoring nutritional status.[3] Moreover, BMI cannot estimate the relative changes in fat and muscle mass occurring in CKD. Body composition analysis with an accurate and sensitive method like dual energy

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absorptiometry (DXA) is often required for accurate assessment of nutritional status.[4,5] DXA results are not affected up to 10% changes in the lean tissue hydration, making it an appropriate tool for assessing body composition in patients without significant volume overload.[6] Even though DXA is often considered as the best method for body composition in predialysis CKD, availability of equipment and high operational costs are major limiting factors, especially when serial measurements are required for assessing the changes in body composition in the outpatient setting. Bioimpedance-based body composition measurements are reported to have less correlation with DXA in predialysis CKD.[7,8] Body composition based on anthropometry is a readily available and less expensive alternative to DXA-based measurements. Anthropometry-based measurements of body fat are reported to be more reliable compared to bioimpedance-based measurements in predialysis CKD.[8,9] There is a relative underestimation of body fat measured by skinfold thickness measurements (SFT) compared to DXA.[8-10] Whether this relative underestimation of body fat remains comparable over a period is not clearly known. In the assessment of nutritional status, longitudinal reproducibility of measurements are more important than a single point measurement. However, there is a lack of longitudinal data on the reliability of anthropometry-based body composition measurements in predialysis CKD compared to DXA. In the current study, we compared the level of agreement between body composition measured with DXA and SFT in CKD Stage 3 and 4, at 2 occasions, 6 months apart. The current study also aims to identify the predictors of the intermethod differences of body fat obtained by DXA and SFT.

Materials and Methods

The data were collected between May 2015 and November 2016 from subjects enrolled to a randomized controlled trial to test the efficacy of bicarbonate supplementation on preserving muscle mass and kidney function. The participants were of Indian ethnicity, aged between 18 and 65 years with CKD Stage 3 and 4 with stable estimated glomerular filtration rate, assessed by Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation. Patients with structural and functional anomalies of the gastrointestinal tract, decompensated chronic liver disease, decompensated heart failure, morbid obesity (BMI $\geq$ 40 kg/m^2), malignancy, chronic infections, prior bicarbonate therapy for a duration of more than 2 weeks, and receiving immunosuppression were excluded. An intention-to-treat analysis showed that the lean body mass improved in the intervention arm, whereas there were no differences in fat mass between 2 arms. A detailed protocol and results of the trial are already published.[10] The body composition at baseline and 6 months was assessed by a DXA scan (Discovery Wi System; Hologic, Bedford, MA). The DXA scans were performed with the subject in the supine position. Rectilinear scans using X-rays were performed from the top of the head, moving toward the feet up to a maximum of 205 lines. A 3-compartment model was used for estimating whole body fat, lean body mass (LBM), and mineral density using QDR software version 13.4.1 (Hologic). A phantom provided by the manufacturer was scanned once a week for quality control. The coefficient of variation was <2% for both lean body mass (LBM) and fat mass, expressed in grams. Patients were clinically euvolemic at the time of DXA scans.

The anthropometric data were recorded by a single trained observer on the day of performing the DXA scans. The SFT was measured at 4 sites (triceps, biceps, subscapular, and iliac crest) on the nondominant side using a slim guide skinfold caliper (Creative Health Products, Ann Arbor, MI). The mean value of 3 consecutive measurements was taken. Body density was calculated using Durnin-Womersley equation. The values were converted to fat percentage using the Siri equation and fat mass was calculated. The instrument was not calibrated in the course of the study. Country-specific definitions were used for the classification of BMI and abdominal obesity.[11]

Statistical Methods

The continuous variables were expressed as mean with 95% confidence intervals (CIs) and categorical variables were expressed as frequencies and percentages. A paired *t*-test was used to assess the agreement between total fat mass assessed by DXA and SFT. The agreements between DXA and SFT were further tested by intraclass correlation coefficients, regression analysis, and Bland-Altman plots. A Student's *t*-test was performed to assess the relationship of selected categorical variables on intermethod differences. A linear regression analysis was performed to identify the patient-related parameters which would account for the

Table 1. Anthropometric and Biochemical Parameters

Parameter	Entry (Mean, 95% CI)	6 Mo (Mean, 95% CI)
Weight (kg)	54.7 (53.12-55.60)	54.13 (52.85-55.41)
Body mass index	21.40 (20.94-21.86)	21.32 (20.84-21.78)
Abdominal circumference (cm)	79.8 (78.8-80.8)	80.2 (79.2-81.2)
Midarm muscle circumference (cm)	22.82 (22.48-23.15)	22.91 (22.59-23.23)
Serum albumin (g/dL)	3.96 (3.90-4.02)	3.95 (3.89-4.01)
Serum cholesterol (mg/dL)	176.41 (170.33-182.48)	177.72 (171.93-183.51)
Hemoglobin (g/dL)	11.81 (11.57-12.04)	11.68 (11.44-11.91)

CI, confidence interval.

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Table 2. Total Body Fat Assessed by DXA and SFT at Baseline and Exit

Group	Baseline			6 Mo		
	Fat Mass in kg (DXA)	Fat Mass in kg (SFT)	*P* value	Fat Mass in kg (DXA)	Fat Mass in kg (SFT)	*P* value
Entire cohort (n = 177)	15.85 (15.07-16.65)	13.71 (13.21-14.32)	<.001	16.13 (15.33-16.93)	13.85 (13.25-14.45)	<.001
Intervention arm (n = 89)	15.94 (14.87-17.02)	13.74 (12.96-14.52)	<.001	16.11 (15.04-17.93)	13.69 (12.86-14.54)	<.001
Control arm (n = 88)	15.76 (14.60-16.91)	13.79 (13.00-14.59)	<.001	16.14 (14.93-17.35)	14.01 (13.15-14.87)	<.001
Males (n = 127)	14.65 (13.85-15.46)	13.24 (12.63-13.85)	<.001	14.79 (14.01-15.58)	13.22 (12.57-13.88)	<.001
Females (n = 50)	18.90 (17.28-20.52)	15.11 (13.98-16.24)	<.001	19.52 (17.80-21.24)	15.46 (14.23-16.67)	<.001
CKD Stage 3 (n = 83)	15.73 (14.54-16.92)	13.57 (12.72-14.31)	<.001	15.72 (14.56-16.88)	13.58 (12.74-14.43)	<.001
CKD Stage 4 (n = 94)	15.95 (14.91-17.00)	13.98 (13.21-14.77)	<.001	16.49 (15.37-17.61)	14.08 (13.23-14.94)	<.001

CKD, chronic kidney disease; DXA, dual-energy absorptiometry; SFT, skinfold thickness.

intermethod differences. A Wilcoxon signed-rank test was used to compare the absolute changes in body fat, which was not normally distributed.

Results

A total number of 188 patients were recruited, among which 177 patients completed the protocol. The data on body composition of 177 patients who completed the study were analyzed for agreement between DXA and SFT.

The baseline mean age of the participants was 50.4 years (95% CI 48.7-52.1). The mean estimated glomerular filtration rate and serum creatinine were 30.6 mL/1.73 m^2 (95% CI 28.9-32.2) and 2.4 mg/dL (95% CI 2.3-2.5), respectively. CKD of unidentified etiology was the predominant cause of CKD. A BMI >23 was present in 27.7% (n = 49) of participants (overweight and obese) and abdominal obesity was present in 68.9% (n = 122). The anthropometric and biochemical parameters at entry and after 6 months are given in Table 1. The lean body mass in the intervention arm assessed by DXA showed an improvement from 36.5 kg (95% CI 35.4-37.8) to 36.8 kg (95% CI 36.5-37.1) over 6 months, whereas the controls showed a decline from 36.2 kg (95% CI 34.9-37.5) to 36.0 kg (95% CI 35.7-36.4), $P = .002$.[10]

Compared to DXA, SFT underestimated the fat mass at baseline as well as 6 months. Similar trends were noted with subgroup analysis for intervention versus controls, gender as well as stage of CKD (Table 2). The intraclass correlation coefficients showed a strong agreement between DXA and SFT at entry as well as exit. The regression and intraclass correlation coefficients at baseline and after 6 months were also similar (Table 3; Fig. 1). Bland-Altman plots also showed a similar degree of agreement between DXA and SFT at baseline and after 6 months (Fig. 2). An XY scatter plot with regression lines for DXA and SFT showed nearly superimposed regression lines, showing that both methods accurately tracked the body fat well over 6 months (Fig. 3). The median change in body fat assessed by DXA over 6 months was 200 g (interquartile range −556 to 974), whereas the changes with SFT were 212 g (IQR −742 to 1373). There were no significant differences in the absolute changes in body fat between DXA and SFT ($P = .517$). The results were similar when the analysis was done after including the patients who withdrew from the study.

The intermethod differences between DXA and SFT at baseline and 6 months were comparable: 2.08 kg (95% CI 1.66-2.5) at baseline versus 2.27 kg (95% CI 1.83-2.71) at 6 months, $P = .200$. The intraclass correlation coefficients for the intermethod differences were 0.866 (95% CI 0.820-0.901). The R^2 value was 0.765. The intermethod differences were higher in females, overweight and obese

Table 3. Intraclass Correlation and Regression Coefficients Between DXA and SFT

Group	Intraclass Correlation Coefficient		Regression Coefficient	
	Baseline	6 Mo	Baseline	6 Mo
Entire cohort (n = 177)	0.894 (0.857-0.921)	0.896 (0.860-0.923)	0.735	0.733
Intervention arm (n = 89)	0.889 (0.831-0.927)	0.876 (0.811-0.919)	0.707	0.696
Control arm (n = 88)	0.898 (0.845-0.933)	0.912 (0.865-0.942)	0.761	0.763
Males (n = 127)	0.903 (0.862-0.931)	0.896 (0.852-0.927)	0.727	0.694
Females (n = 50)	0.892 (0.809-0.938)	0.909 (0.839-0.948)	0.733	0.776
CKD Stage 3 (n = 83)	0.876 (0.808-0.920)	0.885 (0.822-0.926)	0.712	0.696
CKD Stage 4 (n = 94)	0.910 (0.864-0.940)	0.904 (0.855-0.936)	0.763	0.704

CKD, chronic kidney disease; DXA, dual-energy absorptiometry; SFT, skinfold thickness.

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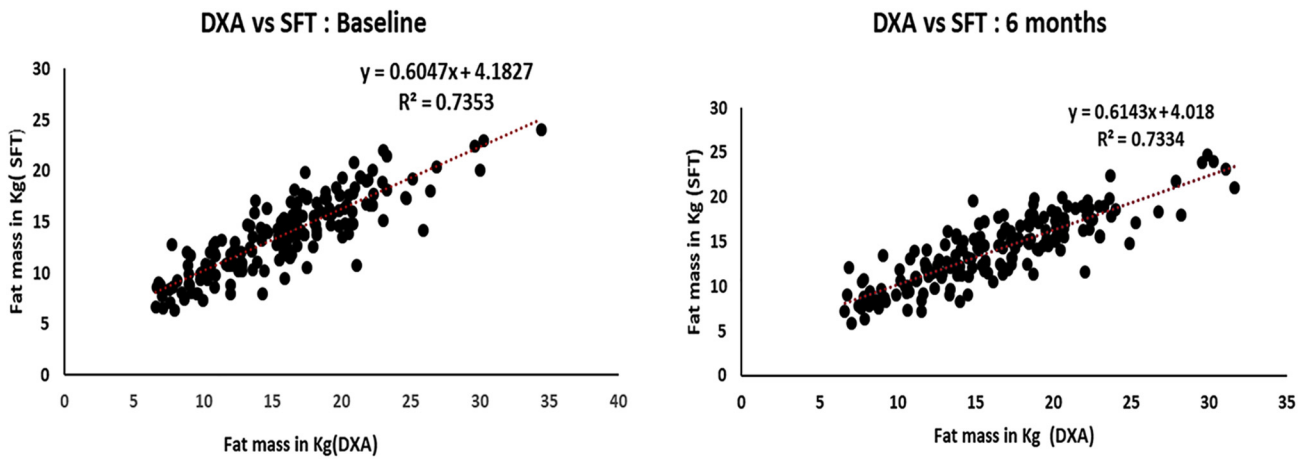

Figure 1. Regression plots showing agreement between DXA and SFT. DXA, dual-energy absorptiometry; SFT, skinfold thickness.

individuals, and those with a higher waist circumference (Table 4).

A linear regression model was created to see the independent influence of gender, age, BMI, and abdominal circumference on the intermethod differences between DXA and SFT at entry and baseline. Age, BMI, and abdominal circumference were analyzed as continuous variables. Gender and BMI turned out to be the significant predictor of intermethod differences at baseline and exit ($P < .001$). Waist circumference and age were not found to influence intermethod differences on linear regression ($P = .677$ and .772 for age; $P = .053$ and .281 for waist circumference).

Discussion

Malnutrition is common in patients with CKD. It is estimated that 20-50% of patients with CKD suffer from PEW.[12] There is no single tool to assess PEW in CKD. BMI and serum albumin levels are the most well-studied indices of nutrition in patients with end-stage renal disease. The International Society of Renal Nutrition and Metabolism recommends using criteria comprising of serum chemistries (albumin, prealbumin, cholesterol), body mass, muscle mass, and dietary intake for diagnosis of PEW.[13] However, BMI has limitations in patients with predialysis CKD. BMI tends to get influenced by changes in the hydration status of the individual and hence might not be reliable. Also, CKD is known to change the fat mass to lean mass ratios. The patient can have normal weight at the expense of having more fat as well as significant loss of lean body mass. Body composition analysis is a much more reliable method for assessing nutritional status in predialysis CKD. A preserved fat mass in the absence of sarcopenia is associated with lesser

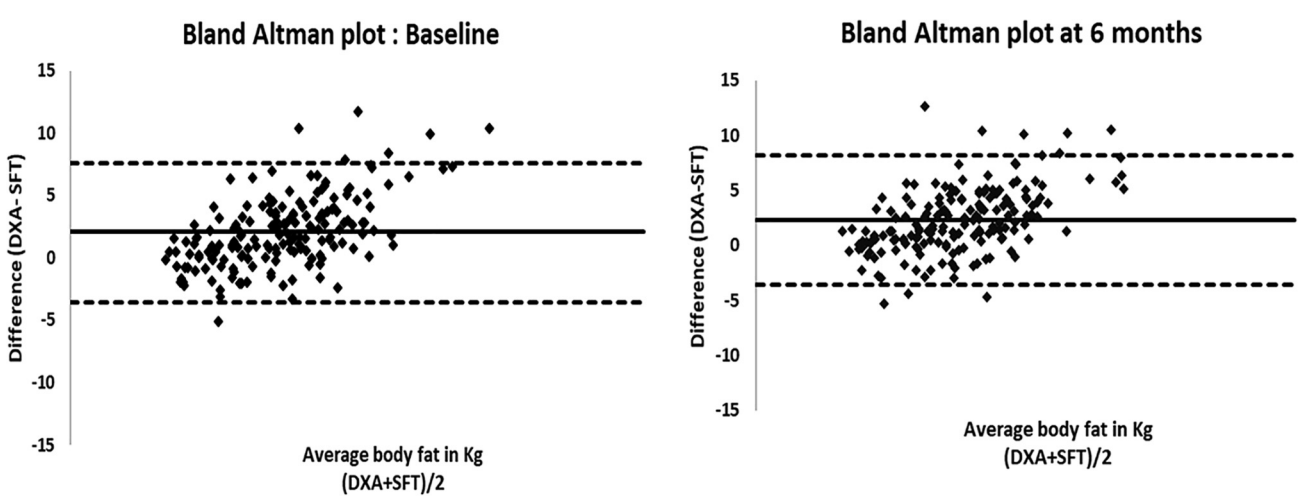

Figure 2. Bland Altman plots showing agreement between DXA and SFT. DXA, dual-energy absorptiometry; SFT, skinfold thickness.

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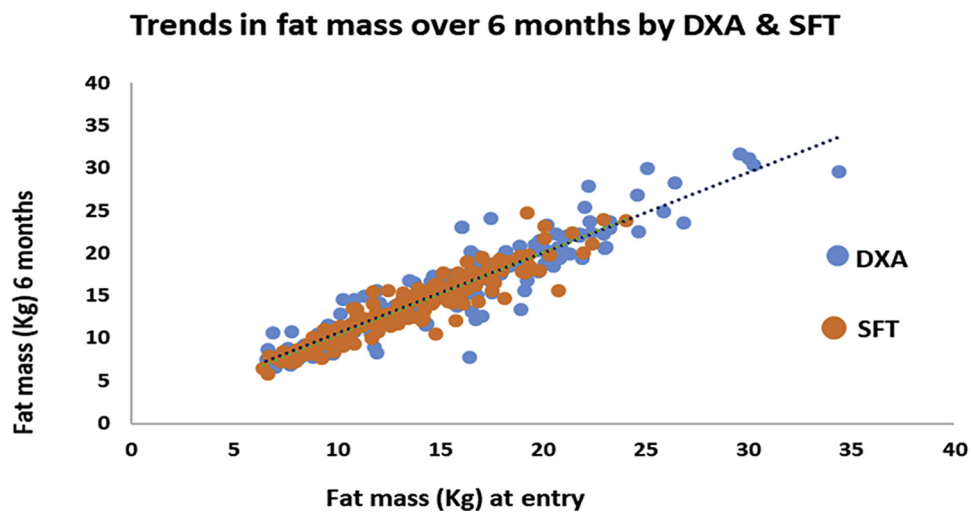

Figure 3. XY scatter plot tracking changes in fat mass with DXA and SFT over 6 months. Dark blue represents regression line for DXA, and green represents regression lines for SFT. DXA, dual-energy absorptiometry; SFT, skinfold thickness. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

mortality rates and progression of disease in predialysis CKD.[2,14] A total body fat of <10% is diagnostic of PEW.[13]

An accurate estimation of body composition might need whole body plethysmography or underwater weighing, which are cumbersome and done only in research settings. DXA is considered as the most reliable nonresearch method for assessing body composition. However, it is expensive and needs dedicated personnel for running and most often will not be available in outpatient care settings. The high operational costs and relative nonavailability of DXA mandate a less expensive, reliable method which could be used in the outpatient settings for serial assessment of body composition to facilitate therapeutic decisions and monitoring to dietary interventions. Bioimpedance assay and SFT-based body composition analysis are 2 easy and less expensive alternatives for assessing body composition in CKD. There are only minimal data on the utility of these methods for assessing body composition in predialysis CKD. Bioimpedance (BIA)-based measurements are reported to overestimate the fat mass and are less reliable compared to SFT-based methods in predialysis CKD.[8,9] SFT-based measurements have shown to have a better agreement compared to BIA in dialysis patients using air displacement plethysmography—the gold standard for assessing body composition.[15] Similarly, SFT-based methods are reported to have a better agreement compared to DXA in kidney transplant recipients as well.[16] There is a lack of longitudinal data on the reliability of anthropometric methods in predialysis CKD.

In the current study, we observed that body fat measured by SFT showed good agreement with the reference method DXA as evidenced by intraclass correlation coefficients, regression, and Bland-Altman plots. The level of agreement was similar when measurements were repeated after 6 months. However, SFT systematically underestimated body fat in the study population, especially in obese individuals and women. Apart from the interoperator

Table 4. Effect of Selected Categorical Variables on the Intermethod Differences

Category	(DXA−SFT) in kg at Entry	*P* value	(DXA−SFT) in kg at 6 Mo	*P* value
Males (n = 127)	1.41 (0.99-1.80)	<.001	1.57 (1.10-2.03)	<.001
Females (n = 50)	3.79 (2.91-4.66)		4.01 (3.20-4.93)	
Normal BMI (n = 128)*	1.4 (0.99-1.82)	<.001	1.73 (1.24-2.21)	<.001
High BMI (n = 49)	3.85 (2.95-4.76)		3.70 (3.83-4.59)	
Normal WC (n = 55)	1.32 (0.62-2.01)	.016	1.37 (0.62-2.11)	.006
High WC (n = 122)†	2.42 (2.15-2.94)		2.68 (2.15-3.22)	
Age >50 (n = 80)	2.06 (1.39-2.74)	.932	2.1 (1.56-2.64)	.659
Age ≤ 50 (n = 97)	2.17 (1.50-2.84)		2.37 (1.77-2.94)	

BMI, body mass index; DXA, dual-energy absorptiometry; SFT, skinfold thickness; WC, waist circumference.
*BMI of 23 taken as cut-off, as per Indian guidelines.
†Waist circumference cut-off: 72 cm in females and 78 cm in males, as per Indian guidelines.

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differences, factors like gender, age, body type, and ethnicity can affect SFT-based measurements. The SFT equations are reported to have high reproducibility in whites compared to other ethnic groups. Body fat equations are derived using regression equations assuming that fat distribution is uniform throughout the body. But, significant ethnic differences are reported in SFT at different sites. People of Asian ethnicity tend to have higher subscapular, abdominal, and iliac crest thickness compared to gender and BMI matched individual from white ethnic groups.[17] For a given BMI, Indians tend to have more fat and lesser lean body mass compared to other races. Indian people tend to have a more nonuniform deposition of fat, especially around abdomen when compared to other ethnic groups.[18] The nonuniform distribution of body fat might affect the performance of the 4 sites in SFT-based equations in Indian population. The existing SFT equations would require a specific correction factor to accommodate the regional variations in fat distribution in persons with Indian ethnicity.

In spite of having a gender correction in the anthropometric equation, SFT is reported to underestimate the fat mass in females with predialysis CKD and end-stage renal disease.[8,17] The DXA-SFT differences in fat mass in the current study was >3.5 kg, which is much higher compared to the previous studies involving patients of non–Indian ethnic origin.[4] Indian women tend to have higher abdominal fat compared to other races.[19] Among the 50 females recruited in the current study, 48 had abdominal obesity. The geographic area where the study was conducted is reported to have a high prevalence of abdominal obesity in females compared to the rest of India. The prevalence of abdominal obesity in the current study is much higher compared to the general population.[20] Age is also reported to be an important factor which might contribute to the intermethod differences between DXA and SFT.[8,18] However, in the current study, we did not find any influence of age on intermethod differences. This might be due to the fact that an upper age limit of 65 was employed for the study.

In the current study, we used a slim guide caliper which is considerably cheaper (USD 13) and has a measurement range of 80 mm. Traditionally, anthropometric measurements are done with Harpenden and Holtain calipers which cost more than 350 USD. Lange caliper costs are approximately 50% lesser and the caliper has a measurement range of 60 mm. The equipment we used is considerably much less expensive and might be useful in low resource settings. The performance of slim guide calipers is reported to be comparable with Harpenden calipers in experimental settings.[21]

To the best of our knowledge, this is the first longitudinal study to assess the reliability of SFT and DXA in predialysis CKD. Most of the previous literature involving single point measurements had shown that SFT underestimates body fat compared to DXA. However, there is only limited information on whether SFT can reliably track the changes in body fat over a period of time. A longitudinal study in kidney transplant recipients has also shown that SFT systematically underestimates body fat by 3-4% compared to DXA, but the degree of underestimation remained similar, when measured multiple times over a period of 1 year.[22] In the current study, SFT appeared to accurately track the changes in body fat over a period of 6 months. SFT-based measurements are prone for significant interobserver differences. To avoid this all the measurements were performed by a single trained observer. The study has a few limitations. We excluded patients with BMI more than 40 because of difficulties in the image acquisition by the DXA scanner. The results of the study cannot be generalized to older individuals with predialysis CKD. All the measurements were done in clinically euvolemic subjects, ascertained by physical examination. However, a few liters of fluid can accumulate without any signs of volume excess. DXA is prone to a small, but consistent error in fat estimation in subjects with overhydration.[23] Even though there was a good reproducibility of SFT compared to DXA, considering the systematic underestimation of body fat, caution should be exercised while using a single SFT value to label patients as malnourished, especially in women, overweight, and obese individuals. However, SFT can be used as a monitoring tool for assessing longitudinal changes in body composition. Considering the differences in fat distribution in Asian races, specific correction factors for gender might improve the performance of existing anthropometry-based equations.

Conclusions

Body fat estimation is a useful tool to diagnose and classify PEW in CKD. A single body fat measurement by SFT has limitations in classifying and diagnosing PEW in subjects of Indian origin with predialysis CKD, as it underestimates total body fat by approximately 2 Kg. However, as the bias is small it can be a useful field method. As there is a good reproducibility of measurements, it is a useful tool for assessing serial changes in body composition over a period. Ethnicity-specific equations need to be developed to overcome the bias associated with SFT.

Practical Applications

SFT-based measurements are a less expensive, practical alternative for nutritional monitoring in predialysis CKD. Even though there is an underestimation of body fat by SFT, the method seems to accurately track the changes over a period of time. Slim guide caliper shows good reproducibility in measurements compared to DXA, making it a suitable tool in resource strained settings.

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